

DFS, BFS ■■ BST – Interview Ready Notes (Hindi)

1. DFS (Depth First Search)

DFS ■■■ ■■ ■■ path ■■ ■■■■ ■■■ explore ■■■■ ■■■, ■■■ ■■■■ ■■■ ■■■■ path ■■■■ ■■■■

DFS Iterative (Stack ■■ ■■■■■):

```
function dfs(root) {  
  if (!root) return [];  
  
  const stack = [root];  
  const values = [];  
  
  while (stack.length > 0) {  
    const node = stack.pop();  
    values.push(node.key);  
  
    if (node.right) stack.push(node.right);  
    if (node.left) stack.push(node.left);  
  }  
  
  return values;  
}
```

2. BFS / Level Order Traversal

BFS ■■■ tree ■■ level by level traverse ■■■■ ■■■■ ■■■

```
function bfs(root) {  
  if (!root) return [];  
  
  const queue = [root];  
  const values = [];  
  
  while (queue.length > 0) {  
    const node = queue.shift();  
    values.push(node.key);  
  
    if (node.left) queue.push(node.left);  
    if (node.right) queue.push(node.right);  
  }  
  
  return values;  
}
```

3. Binary Search Tree (BST)

BST ■■■ left subtree ■■■■ ■■ right subtree ■■■■ ■■■■ ■■■■

```
class BstNode {
    constructor(key) {
        this.key = key;
        this.left = null;
        this.right = null;
    }
}

class BinarySearchTree {
    constructor() {
        this.root = null;
    }
}

Insert Operation:
insert(key) {
    const newNode = new BstNode(key);
    if (!this.root) {
        this.root = newNode;
    } else {
        this.insertNode(this.root, newNode);
    }
}
```